SQL Query Pizza Sales Dashboard SQL+BI project

--**Create database**

Create database SQL_PowerBI_Project4_pizza_sales

Use SQL_PowerBI_Project4_pizza_sales

--Import CSV file to SSMS (Steps uploaded in the separate file)

--this fie creates table called dbo.pizza_sales

--Data Cleaning

```
Select * from [dbo].[pizza_sales]
Where [pizza_id] is null
 or  [order_id] is null
 or [pizza_name_id] is null
 or [pizza_category]  is null
 or [quantity] is null
 or [order_date] is null
 or [order_time] is null
 or [unit_price] is null
 or  [total_price] is null
 or [pizza_size] is null
 or [pizza_ingredients] is null
 or [pizza_name] is null;
```

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Results | Messages

pizza_id	order_id	pizza_name_id	quantity	order_date	order_time	unit_price	total_price	pizza_size	pizza_category	pizza_ingredients	pizza_name

--Checking for duplicates

```
Select * from [dbo].[pizza_sales]

Select [pizza_id],  [order_id], count(*) as count_order_id
from [dbo].[pizza_sales]
group by [pizza_id] , order_id
having count(*) > 1;
```

Results | Messages

pizza_id	order_id	count_order_id

--Data Transformation

/*Data Transformation - pizza size re presented as S, M, L, XL,XXL in data,

we need to transform it to S= small, M= Medium, L=Large, XL = XLarge , XXL = XXLarge)*/

--we can do the transformations in Power BI

---KPI requirements

```sql
--Total Revenue - The sum of the total price of all pizza orders

Select sum([total_price]) as total_revenue from[dbo].[pizza_sales]
```

	total_revenue
1	817860.049999993

```sql
--Average Order Value - The average amount spent per order, calculated by dividing the total revenue by the total number of orders

Select sum([total_price]) / count(distinct [order_id]) as avg_order_value from [dbo].[pizza_sales]
```

Results Messages

	avg_order_value
1	38.3072622950816

```sql
--Total Pizzas Sold - The sum of the quantities of all pizzas sold

Select sum([quantity]) as total_pizza_sold from [dbo].[pizza_sales] --- it throws error so we change data type to int


alter table [dbo].[pizza_sales]
alter column [quantity] int not null;


Select sum([quantity]) as total_pizza_sold from [dbo].[pizza_sales]
```

	total_pizza_sold
1	49574

```sql
--Total Orders - The total number of orders placed.

Select Count(distinct [order_id]) as Total_Orders from [dbo].[pizza_sales]
```

Results Messages

	Total_Orders
1	21350

```sql
--Average Pizzas Per Order - The average number of pizzas sold per order,
--calculated by dividing the total number of pizzas sold by the total number of orders.

Select sum([quantity]) /Count(distinct [order_id]) as avg_pizza_per_order from [dbo].[pizza_sales]
```

	avg_pizza_per_order
1	2

--chart requirements

```sql
-- Daily Trend For Total Orders - Create a bar chart that displays the daily trend of total orders
--over a specific time period. This chart will help us identify any patterns or fluctuations in the order volumes on a daily basis.
/*
we need to first find the day of weeks as order_day then count the orders
*/
Select DATENAME(DW, [order_date]) as Order_day, count(distinct [order_id]) as total_orders
From [dbo].[pizza_sales]
group by DATENAME(DW, [order_date]);
```

	Order_day	total_orders
1	Saturday	3158
2	Wednesday	3024
3	Monday	2794
4	Sunday	2624
5	Friday	3538
6	Thursday	3239
7	Tuesday	2973

```sql
-- Monthly Trend for Total Orders - Create a line chart that illustrates the hourly trend of total orders throughout the day.
--This chart will allow us to identify peak hours or periods of high order activity.

/*
we need to first find the day of weeks as order_month then count the orders
*/

Select DATENAME(Month, [order_date]) as Order_month, count([order_id]) as total_orders
from [dbo].[pizza_sales]
Group by DATENAME(Month, [order_date]);
```

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	Order_month	total_orders
1	February	3892
2	June	4025
3	August	4094
4	April	4067
5	May	4239
6	December	3859
7	January	4156
8	September	3819
9	October	3797
10	July	4301
11	November	4185
12	March	4186

```sql
-- Percentage of Sales by Pizza Category - Create a pie chart that shows the distribution
--of sales across different pizza categories. This chart will provide insights into the
--popularity of various pizza categories and their contribution to overall sales.

Select [pizza_category],
    Round(Sum([total_price]), 2) as total_revenue,
    Round(Sum([total_price])* 100 / (Select sum([total_price])from [dbo].[pizza_sales]),2) as percentage_of_pizza_sales_by_category
FROM [dbo].[pizza_sales]
Group by [pizza_category];
```

Results | Messages

	pizza_category	total_revenue	percentage_of_pizza_sales_by_category
1	Classic	220053.1	26.91
2	Chicken	195919.5	23.96
3	Veggie	193690.45	23.68
4	Supreme	208197	25.46

```sql
-- Percentage of Sales by Pizza Size - Generate a pie chart that represents
--the percentage of sales attributed to different pizza sizes. This chart will help us understand
--customer preferences for pizza sizes and their impact on sales.

Select [pizza_size],
    Round(Sum([total_price]), 2) as total_revenue,
    Round (Sum([total_price]) *100 / (Select SUM([total_price]) from[dbo].[pizza_sales]),2) as Percentage_of_pizza_sales_by_size
FROM [dbo].[pizza_sales]
Group by [pizza_size];
```

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Results | Messages

	pizza_size	total_revenue	Percentage_of_pizza_sales_by_size
1	L	375318.7	45.89
2	XXL	1006.6	0.12
3	M	249382.25	30.49
4	XL	14076	1.72
5	S	178076.5	21.77

```sql
-- Total Pizzas Sold by Pizza Category - Create a funnel chart that presents
--the total number of pizzas sold for each pizza category. This chart will allow us to compare
--the sales performance of different pizza categories.

Select [pizza_category], sum([quantity]) as total_quantity_of_pizza_sold
From [dbo].[pizza_sales]
Group by [pizza_category]
Order by total_quantity_of_pizza_sold;
```

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Results Messages

	pizza_category	total_quantity_of_pizza_sold
1	Chicken	11050
2	Veggie	11649
3	Supreme	11987
4	Classic	14888

```sql
-- Top 5 Best Sellers by Revenue, Total Quantity, and Total Orders
--- Create a bar chart highlighting the top 5 best-selling pizzas based on the Revenue,
--Total Quantity, Total Orders. This chart will help us identify the most popular pizza options.

Select top 5 [pizza_name], sum([total_price]) as total_revenue, count([order_id]) as top5_total_orders
from [dbo].[pizza_sales]
group by [pizza_name]
order by total_revenue DESC;
```

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Results Messages

	pizza_name	total_revenue	bottom5_total_orders
1	The Brie Carre Pizza	11588.4999999999	480
2	The Green Garden Pizza	13955.75	987
3	The Spinach Supreme Pizza	15277.75	940
4	The Mediterranean Pizza	15360.5	923
5	The Spinach Pesto Pizza	15596	957

```sql
---top 5 pizza by quantity

Select top 5 [pizza_name], sum([quantity]) as top5_total_quantity_sold
from [dbo].[pizza_sales]
group by [pizza_name]
order by top5_total_quantity_sold DESC;
```

Results | Messages

	pizza_name	top5_total_quantity_sold
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

```sql
---bottom 5 pizza by quantity

Select top 5 [pizza_name], sum([quantity]) as bottom5_total_quantity_sold
from [dbo].[pizza_sales]
group by [pizza_name]
order by bottom5_total_quantity_sold ASC;
```

123 %

Results | Messages

	pizza_name	bottom5_total_quantity_sold
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Pizza	950
5	The Soppressata Pizza	961

```sql
---top 5 pizza by total order

Select top 5 [pizza_name], count(distinct [order_id]) as top5_total_no_of_order
from [dbo].[pizza_sales]
group by [pizza_name]
order by top5_total_no_of_order DESC;
```

	pizza_name	top5_total_no_of_order
1	The Classic Deluxe Pizza	2329
2	The Hawaiian Pizza	2280
3	The Pepperoni Pizza	2278
4	The Barbecue Chicken Pizza	2273
5	The Thai Chicken Pizza	2225

```
---bottom 5 pizza by total order

Select top 5 [pizza_name], count(distinct [order_id]) as bottom5_total_no_of_order
from [dbo].[pizza_sales]
group by [pizza_name]
order by bottom5_total_no_of_order ASC;
```

123 %

Results Messages

	pizza_name	bottom5_total_no_of_order
1	The Brie Carre Pizza	480
2	The Mediterranean Pizza	912
3	The Spinach Supreme Pizza	918
4	The Calabrese Pizza	918
5	The Chicken Pesto Pizza	938

---Below queries are for concept purpose, we will not include it in our Power BI report.

```sql
/*Top 5 Classic pizzas of Medium (M) or Large (L) size that have been ordered more than once,
sorted by the total number of distinct orders in ascending order. */

/*
The query will return a table with four columns (pizza_name, pizza_category, pizza_size, and Total_Orders)
showing the 5 least ordered Medium or Large 'Classic' pizzas that appeared in more than one distinct order.*/

SELECT Top 5  pizza_name,[pizza_category],[pizza_size], COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales
WHERE pizza_category = 'Classic' and [pizza_size] in ('M', 'L')
GROUP BY pizza_name, [pizza_category],[pizza_size]
Having COUNT(DISTINCT order_id) > 1
ORDER BY Total_Orders ASC;
```

123 %

Results | Messages

	pizza_name	pizza_category	pizza_size	Total_Orders
1	The Greek Pizza	Classic	L	255
2	The Greek Pizza	Classic	M	279
3	The Pepperoni, Mushroom, and Peppers Pizza	Classic	L	381
4	The Pepperoni, Mushroom, and Peppers Pizza	Classic	M	391
5	The Italian Capocollo Pizza	Classic	M	399

```sql
---Bottom 5 "Classic" category pizzas in sizes Medium ('M') and Large ('L'),
--ranked by the total number of unique orders for each, but only includes those pizzas
--- that have been ordered at least twice.
SELECT Top 5 pizza_name,[pizza_category],[pizza_size], COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales
WHERE pizza_category = 'Classic' and [pizza_size] in ('M', 'L')
GROUP BY pizza_name, [pizza_category],[pizza_size]
Having COUNT(DISTINCT order_id) > 1
ORDER BY Total_Orders DESC;
```

Results | Messages

	pizza_name	pizza_category	pizza_size	Total_Orders
1	The Classic Deluxe Pizza	Classic	M	1159
2	The Pepperoni Pizza	Classic	M	918
3	The Hawaiian Pizza	Classic	L	896
4	The Italian Capocollo Pizza	Classic	L	715
5	The Pepperoni Pizza	Classic	L	712